

GNU nano 7.2

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <ctype.h>

void expand_variables(char *input, char *output, int size)
{
    int i = 0;
    int j = 0;

    while (input[i] != '\0' && j < size - 1)
    {
        if (input[i] == '$')
        {
            i++;

            char variable[100];
            int k = 0;

            while (isalnum(input[i]) || input[i] == '_')
            {
                if (k < 99)
                    variable[k++] = input[i];

                i++;
            }

            variable[k] = '\0';

            char *value = getenv(variable);

            if (value != NULL)
            {
                int v = 0;

                while (value[v] != '\0' && j < size - 1)
                {
```

```
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        while (value[v] != '\0' && j < size - 1)
        {
            output[j++] = value[v++];
        }
    else
    {
        printf("Undefined variable: %s\n", variable);
    }
}
else
{
    output[j++] = input[i++];
}
}

output[j] = '\0';
}

int main()
{
    char input[500];
    char output[1000];

    printf("Enter a string: ");
    fgets(input, sizeof(input), stdin);

    input[strcspn(input, "\n")] = '\0';

    expand_variables(input, output, sizeof(output));

    printf("Expanded string: %s\n", output);

    return 0;
}
```

```
sravya@Sravya1955: ~/os
sravya@Sravya1955:~$ cd ~/os
sravya@Sravya1955:~/os$ nano variable_expand.c
sravya@Sravya1955:~/os$ gcc variable_expand.c -o variable_expand
sravya@Sravya1955:~/os$ ./variable_expand
Enter a string: hello
Expanded string: hello
sravya@Sravya1955:~/os$ ./variable_expand
Enter a string: My home is $HOME
Expanded string: My home is /home/sravya
sravya@Sravya1955:~/os$ ./variable_expand
Enter a string: PATH is $PATH
Expanded string: PATH is /home/sravya/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/WINDOWS/system32:/mnt/c/WINDOWS:/mnt/c/WINDOWS/System32/Wbem:/mnt/c/WINDOWS/System32/WindowsPowerShell/v1.0:/mnt/c/WINDOWS/System32/Op
enSSH:/mnt/c/Program Files/dotnet:/mnt/c/Program Files/PuTTY:/mnt/c/Users/chand/AppData/Local/Programs/Python/Launcher:/mnt/c/Users/chand/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/chand/Downloads/38833FF268A1D.UnigramPreview_g9c9v27vpyspw!App/gcc_15_1_0_gdb_16_3_90_20250511_binutils_2_44_mingw_w64_v1
2_0_0/bin:/mnt/d/Users/chand/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Program Files/Java/jdk-21/bin:/mnt/c/Users/chand/AppData/Local/Programs/Mic
rosoft VS Code/bin:/snap/bin
sravya@Sravya1955:~/os$ ./variable_expand
Enter a string: My name is $XYZ123
Undefined variable: XYZ123
Expanded string: My name is
sravya@Sravya1955:~/os$
```

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>

void pwd_command()
{
    char cwd[1024];

    if (getcwd(cwd, sizeof(cwd)) != NULL)
        printf("%s\n", cwd);
    else
        perror("pwd");
}

void cd_command(char *path)
{
    if (path == NULL)
    {
        printf("cd: missing argument\n");
        return;
    }

    if (chdir(path) != 0)
        perror("cd");
}

void echo_command(char *text)
{
    if (text != NULL)
        printf("%s\n", text);
}

int main()
{
    char input[1024];
```

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```
        cd_command(input + 3);
    }
    else if (strncmp(input, "echo ", 5) == 0)
    {
        echo_command(input + 5);
    }
    else if (strcmp(input, "exit") == 0)
    {
        printf("Exiting shell...\n");
        break;
    }
    else
    {
        printf("Invalid command: %s\n", input);
    }
}

return 0;
}
```

Expanded string: My name is

sravya@Sravya1955:~/os\$

sravya@Sravya1955:~/os\$ nano builtin_shell.c

sravya@Sravya1955:~/os\$ nano builtin_shell.c

sravya@Sravya1955:~/os\$ gcc builtin_shell.c -o builtin_shell

sravya@Sravya1955:~/os\$./builtin_shell

myshell> pwd

/home/sravya/os

myshell> echo Hello Sravya

Hello Sravya

myshell> cd ..

myshell> pwd

/home/sravya

myshell> cd os

myshell> ^C

sravya@Sravya1955:~/os\$./builtin_shell

myshell> hello

Invalid command: hello

myshell> exit

Exiting shell...

sravya@Sravya1955:~/os\$ |

